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GMRIT/ADM/F-03
REV.: 03

COURSE HANDOUT
B. Tech. – 3rd Semester

Course Title : **Data Structures**
Course Code : **21CS303**
Course Structure : **3-0-0-3**
Course coordinator : **Dr. S. Akila Agnes**
Instructor(s) : **Dr. V. Vasudha Rani, Dr. Satish M, Dr.V. Srinadh, Dr. S. Akila Agnes, Mrs. M. Sravani, Mr. G. Dharma Raju, Ms. P. Pooja**

Dated: 14.08.2023
Academic Year: 2023-2024

Course Description : Data structures provide specialized methods for efficient data storage and organization within computer systems. These structures are essential for designing optimized algorithms. Numerous programming languages offer functions for storing and retrieving values, necessitating an understanding of linear and non-linear data structures. Linear data structures, such as stacks, queues, linked lists, and hashing, optimize data storage and retrieval. Units I and II focus on linear data structures, providing insights into handling sequentially arranged data that can be traversed in a single pass. Hierarchical relationships among data items often require representation. Units III and IV delve into non-linear data structures like trees, binary search trees, heaps, balanced trees, B-trees, and graphs. These structures effectively represent hierarchical relationships among data elements.

Scope and Objective:

- Understanding data structures and their roles in efficient storage and organization informs algorithm design.
- Implement and use linear (stacks, queues, linked lists, hashing) and non-linear (trees, graphs) data structures to improve data manipulation and hierarchical representation.

Textbook (s)

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C, Pearson Education, 2002
2. Michael Main, Walter Savitch, Data Structures, and other objects using C++, 4th Edition, Addison Wesley, 2018

Reference (s)

1. S. Tanenbaum, Y. Langsam and M.J. Augenstein,, Data Structures using C and C++, 2nd Edition, Pearson Education, 2015
2. 2. R. F. Gilberg, B. A. Forouzan, Data Structures A Pseudocode Approach with C, 2nd Edition, CENGAGE Learning, 2005

Course Outcomes

At the end of the Course, students will be able to:

1. Describe the operations and implementation of List ADT (Understand)
2. Comprehend the operations and implementation of Stack and Queue (Understand)
3. Illustrate the applications of linear data structures (Apply)
4. Describe the operations and implementation of hash table (Understand)
5. Comprehend the operations and implementation of tree data structure (Understand)
6. Illustrate the variations of tree data structure (Understand)

COs – POs Mapping

COs	PO_1	PO_2	PO_{12}
1	3	2	1
2	3	2	1
3	2	3	1
4	2	3	1
5	3	2	1
6	2	3	1

3–Strongly | 2–Moderately linked | 1–Weakly linked

SYLLABUS:

Unit I

Linear Data Structures – List **12 Hours**

Abstract Data Types (ADTs) – List ADT – Array-based implementation – Linked list implementation –Singly linked lists- Circularly linked lists- Doubly-linked lists – Applications of lists –Polynomial Manipulation – All operations (Insertion, Deletion, Merge, Traversal) - Searching - Linear Search – Binary Search

Doubly linked Circular list

Unit II

Linear Data Structures – Stack and Queue **12 Hours**

Stack ADT – Array implementation – Linked list implementation – Applications of Stack – infix to postfix conversion, evaluation of postfix expression – Queue ADT – Array implementation – Linked list implementation – Application of Queue – Ticket counter

Circular Queue

Unit III

Sorting, Hashing Techniques and Trees **12 Hours**

Sorting – Bubble Sort – Selection Sort – Insertion Sort – Shell Sort – Radix Sort – Quick Sort – Merge Sort - Hashing - Hash Functions – Separate Chaining – Open Addressing – Rehashing

Trees: Introduction, Terminology, Binary Trees, Representation of Binary Trees using arrays and linked lists, Binary tree traversals

Extendible Hashing

Unit IV

Variations on Trees and Graphs **12 Hours**

Binary Search Trees: definition, basic operations of BST (Searching, Insertion and deletion) - Introduction to AVL trees: Balancing AVL tree by rotations after insertions and deletions of a data node multi-way search trees: Introduction to m-way search trees, B-trees, B+ Trees;

Heaps: Binary heaps, definition of a Max-heap, Min-heap, Creating Max-Heap, Applications: Heap sort Graphs: Terminology, Representation, Traversals: Depth First Search and Breadth First Search

Priority queue operations: insertions and extract-max

Total: 48 Hours

Course Plan:

Lecture No.	Learning objectives	Topic(s) to be covered	Textbook/ Reference
		UNIT – I	
1	Introduction to ADT	Abstract Data Types (ADTs), List ADT	T1, R1
2	To study the arrangement of elements in LIST, Implementation of list using arrays	Array-based implementation	T1, R1
3	Discussion of Implementation of list using linked list	Linked list implementation	T1, R1
4	To study the arrangement of elements in single linked list fashion	Singly linked lists	T1, R1
5			
6	To study the arrangement of elements in a Doubly Linked List	Doubly Linked Lists	T1, R1
7	To study the arrangement of elements in a Circular Linked List	Circular linked lists	T1, R1
8	Applications of lists	Applications of lists	T1, R1
9	Applications of lists	Polynomial Manipulation	T1, R1
10	Polynomial Manipulation	All operations (Insertion, Deletion, Merge, Traversal)	T1, R1
11	To Search any given key in the given sequence and To compare Binary Search with Linear search	Searching - Linear Search	T1, R1
12		Binary Search	T1, R1
		UNIT – II	
13	To understand use of stack in recursion	Stack ADT	T1, R1
14	To understand implementation of stacks using Arrays	Array implementation	T1, R1
15	To perform Operations on Singly Linked Lists	Linked list implementation	T1, R1
16	To understand implementation of stacks using Dynamic memory allocation	Applications of Stack	T1, R1
17		Infix to postfix conversion	T1, R1
18	To understand implementation of stacks using Dynamic memory allocation	Evaluation of postfix expression	T1, R1
19	To study the arrangement of elements in a Queue	Queue ADT	T1, R1
20	To understand implementation of queue using arrays	Array implementation	T1, R1
21	To understand implementation of queue using linked list	Linked list implementation	T1, R1
22	To understand about Applications of Queue	Applications of Queue	T1, R1
23		Ticket counter	T1, R1
24		Ticket counter	T1, R1

	UNIT – III		
25	To sort given elements using Bubble, Selection and Insertion sort methods and compare these sorting methods	Sorting	T1, R1
26	To sort given elements using Bubble, Selection and Insertion sort methods and compare these sorting methods	Bubble Sort	T1, R1
27		Selection Sort	T1, R1
28		Insertion Sort	T1, R1
29	To sort given elements using Shell, Radix sorting methods	Shell Sort	T1, R1
30		Radix sort	T1, R1
31	To sort given elements Quick and merge sort methods and compare these sorting methods	Quick Sort	T1, R1
32		Merge Sort	T1, R1
33	Significance of hashing over linear & linked lists; Methods of hashing	Hashing: Hash Functions, Separate Chaining	T1, R1
34		Open Addressing, Rehashing	T1, R1
35	Implementation of BT using dynamic memory allocation	Trees: Introduction- Terminology - Binary Trees - Representation of Binary Trees using arrays and linked lists	T1, R1
36	Different orders of visiting nodes in the BT (in, pre and post orders)	Binary tree traversals	T1, R1
	UNIT – IV		
37	To delete an element from BT, Comparison of BT&BST; To search an element in BT.	Binary Search Trees: definition - basic operations of BST (Searching, Insertion and deletion	T1, R1
38	Significance of AVL Tree and its height	Introduction to AVL trees	T1, R1
39	To reduce height using rotations	Height of an AVL Tree - Balancing AVL tree by rotations after insertions of a data node	T1, R1
40	To reduce height using rotations	Height of an AVL Tree - Balancing AVL tree by rotations after deletions of a data node	T1, R1
41	To construct different trees with order M	Multi-way search trees: introduction to m-way search trees	T1, R1
42	Introduction to B+ tree	B-trees, B+ Trees	T1, R1
43	Introduction to Heaps	Heaps: Binary Heaps	T1, R1
44	To insert and delete elements in Heap	definition of a Max-heap , Min-heap, creating Max-Heap	T1, R1
45	Applications of Heap	Applications: Heap sort	T1, R1
46	Introduction to Graphs	Graph Terminologies, Representation	T1, R1
47	Graph Traversal	Depth First Search	T1, R1
48	Graph Traversal	Breadth First Search	T1, R1

Internal Assessment Pattern

Cognitive Level	Int. Test 1 (%)	Int. Test 2 (%)	Open Book Exam (%)
Remember	20	20	-
Understand	60	60	-
Apply	20	20	80
Analyze	-	-	20
Evaluate	-	-	-
Create	-	-	-
Total (%)	100	100	100

Evaluation Scheme

Component	Duration (minutes)	Marks	% *Weightage*	Date & Time	Venue
Internal Test-1	90	40	20 (80% of Best + 20% of Rest)	9.10.2023 to 11.10.2023	1st Block / 5th Block
Internal Test-2	90	40		07.12.2023 to 13.12.2023	1st Block / 5th Block
Open Book Exam / Assignment Test	120	30	10	07.12.2023 to 13.12.2023	1st Block / 5th Block
Semester End Exams	180	70	70	25.12.2023 to 06.01.2024	1/2/3/4/5/6 Blocks

Chamber Consultation Hour: 4.00PM

Venue: CSE / IT Staff Room

Notices: CSE/IT Notice Board

Course Coordinator

Dr. S. Akila Agnes, Sr.Asst. Prof/ CSE

Course Instructors

Dr. V. Vasudha Rani, Assoc. Prof/ IT

Dr. V. Srinadh, Assoc. Prof / CSE-AIML

Dr. Satish M, Assoc. Prof / CSE

Dr. S. Akila Agnes, Sr. Asst. Prof/ CSE

Mr. G. Dharma Raju, Asst. Prof/ CSE

Mrs. M. Sravani, Asst. Prof/ CSE

Ms. P.Pooja, Asst. Prof/ IT

HOD/CSE